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Real-world Dosing Patterns of Biologics in Crohn’s Disease and Ulcerative Colitis: A Retrospective Observational Study

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OBJECTIVE

- To evaluate dosing patterns across biologics in Crohn’s disease (CD) and ulcerative colitis (UC) in the United States (US).

CONCLUSIONS

- This study in a US population showed dose escalation was common but occurred at varying rates between biologic treatments.
- While CD and UC are distinct diseases, the dose escalation observed for each biologic was generally similar across the two conditions.
- The discontinuation rates were higher among non-escalators, and higher rates of switching were observed among escalators who discontinued.

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BACKGROUND

- Crohn’s disease (CD) and ulcerative colitis (UC) are chronic inflammatory conditions with varying severity and a relapsing-remitting disease course.^{1,2}
- Biologics have become mainstay treatments for moderately to severely active CD and UC; however, patients may lose responsiveness over time requiring therapy change.³
- Evaluating treatment patterns across biologics can aid decision-makers in optimizing therapeutic management for patients with CD or UC.⁴

RESULTS

Patient demographics and clinical characteristics

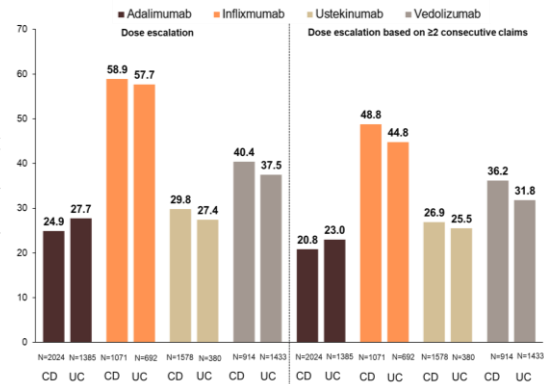
Characteristics	CD cohort (N=6,060)	UC cohort (N=4,556)
Age median (IQR) (years)	40.0 (28.0–52.0)	42.0 (31.0–54.0)
Sex, n (%)		
Female	3190 (52.6)	2133 (46.8)
Median total duration of treatment (months)	20.5	19.4
Median duration of follow-up (months)	26.4	25.0
Charlson Comorbidity Index (CCI), Mean (SD)	0.52 (1.0)	0.45 (1.0)
CCI comorbidities*, n (%)		
Chronic pulmonary disease	703 (11.6)	486 (10.7)
Diabetes	475 (7.8)	393 (8.6)
Liver disease	381 (6.3)	220 (4.8)
Other comorbidities, n (%)		
Anxiety or depression	1693 (27.9)	1074 (23.6)
Obesity	928 (15.3)	687 (15.1)
Asthma	518 (8.5)	358 (7.9)
Baseline treatments, n (%)		
Corticosteroids	4092 (67.5)	3745 (82.2)
Immunomodulators	2718 (44.9)	3644 (80.0)
Previous biologics	2420 (39.9)	1762 (38.7)
Previous surgery†	605 (10.0)	29 (0.6)
Previous other procedures††	5210 (86.0)	3774 (82.8)

Note: *Indicates the most common CCI comorbidities for CD and UC. There were 17 CCI conditions in total. Data for diabetes included mild or moderate and diabetes with chronic complications; Liver disease included mild and moderate or severe liver disease.
†Previous surgery included bowel resection, fistula removal, ileal pouch-anal anastomosis, ileostomy, and stricturoplasty.
††Previous other procedures included abdominal computed tomography, abdominal/pelvic magnetic resonance imaging, colonoscopy, esophagogastroduodenoscopy, and sigmoidoscopy.

- Overall, the proportion of patients with claims indicating the induction phase was 91.7% for CD and 95.7% for UC across all biologics.
- Overall, the proportion of patients with claims indicating initiation of maintenance was 97.3% for CD and 96.0% for UC across all biologics.

Abbreviations
CCI, Charlson Comorbidity Index; CD, Crohn’s disease; IQR, interquartile range; N, total number of patients; n, number of patients in the specified category; SD, standard deviation; UC, ulcerative colitis; US, United States.
References
1. Torres, J, et al. *Lancet*. 2017;389:1741-1755.
2. Ordás, Ingrid et al. *Lancet*. 2012;380:1606-1619.
3. Khan, Shahnaz, et al. *J Clin Phar Ther*. 2019;44:495-507.
4. Dovizio, Melania, et al. *Adv Ther*. 2024;41:1-17.
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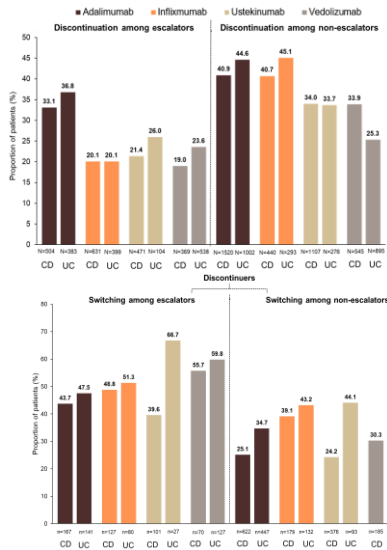
Similar rates of dose escalation were observed between CD and UC for the two escalation definitions across treatments



Sensitivity analysis showed similar rates of dose escalation in CD and UC across all biologics

Note: N is the number of patients with an evidence of maintenance phase.

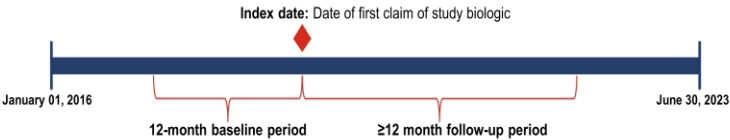
Discontinuation rates were higher among non-escalators, while higher rates of switching were observed among escalators who discontinued



Note: N is the number of patients with or without dose escalation and n is the number of patients who discontinued index treatment (for switching).

METHODS

- Study design:** A retrospective, observational study
- Data source:** Merative™ MarketScan® Commercial and Medicare Databases
- Biologics:** Adalimumab, infliximab, ustekinumab, vedolizumab



Study cohorts	
CD	UC
• ≥1 inpatient or non-diagnostic outpatient claim with a diagnosis of CD.	• ≥1 inpatient or non-diagnostic outpatient claim with a diagnosis of UC.
• No inpatient or non-diagnostic outpatient claims with a diagnosis of UC	• No inpatient or non-diagnostic outpatient claims with a diagnosis of CD
• No evidence during the study, of other selected autoimmune conditions: ankylosing spondylitis, atopic dermatitis, axial spondyloarthritis, hidradenitis suppurativa, juvenile idiopathic arthritis, psoriasis, uveitis, psoriatic arthritis, rheumatoid arthritis.	
Assessments	
Dose escalation	≥20% increase in the starting maintenance dose or early administration (≥7 days early)
• Sensitivity analysis for dose escalation	≥2 consecutive claims with ≥20% dose increase in the starting maintenance dose or early administration (≥7 days early)
Discontinuation	≥90-day gap in days of supply of index biologic
• Switching	≥1 claim for a non-index biologic or Janus kinase inhibitor among discontinuers
▷ Data were reported using descriptive statistics for categorical variables (n, mean, standard deviation, frequency).	